

Crew MultiAgent

CrewAI Multi-Agent Example: Complete Marketing Team

Multiple specialized agents working together on marketing campaigns.

"""

```
from crewai import Agent, Task, Crew, Process
from langchain_openai import ChatOpenAI
from langchain.tools import Tool
from typing import List
```

```
# Initialize LLM
```

```
llm = ChatOpenAI(
    model="gpt-4",
    temperature=0.7,
    api_key="your-api-key-here"
)
```

```
# Define shared tools
```

```
def market_research_tool(topic: str) -> str:
```

```
    """Market research tool"""
```

```
    return f"Market research data for {topic}: [Demographics, trends, competitors]"
```

```
def seo_analysis_tool(content: str) -> str:
```

```
    """SEO analysis tool"""
```

```
    keywords = ["AI", "automation", "productivity", "technology"]
```

```
    return f"SEO Analysis: Keywords found: {keywords}, Score: 78/100"
```

```
def social_metrics_tool(platform: str) -> str:
    """Social media metrics"""
    return f"{platform} metrics: Engagement rate: 4.2%, Reach: 50K, Best time: 2PM"

def competitor_analysis_tool(company: str) -> str:
    """Competitor analysis"""
    return f"Competitor {company}: Strengths, weaknesses, market position analysis"

# Create tools
research_tool = Tool(
    name="MarketResearch",
    func=market_research_tool,
    description="Conduct market research on any topic or industry"
)

seo_tool = Tool(
    name="SEOAnalysis",
    func=seo_analysis_tool,
    description="Analyze content for SEO optimization"
)

social_tool = Tool(
    name="SocialMetrics",
    func=social_metrics_tool,
    description="Get social media metrics and insights"
)

competitor_tool = Tool(
```

```

    name="CompetitorAnalysis",
    func=competitor_analysis_tool,
    description="Analyze competitors and market positioning"
)

# Agent 1: Market Research Analyst
market_analyst = Agent(
    role="Market Research Analyst",
    goal="Provide deep market insights and identify target audiences",
    backstory="""You are an experienced market research analyst with expertise in:
    - Consumer behavior analysis
    - Market segmentation
    - Competitive intelligence
    - Trend forecasting

    You excel at turning data into actionable insights that drive strategy.""",
    tools=[research_tool, competitor_tool],
    llm=llm,
    verbose=True,
    allow_delegation=False
)

# Agent 2: Content Strategist
content_strategist = Agent(
    role="Content Strategist",
    goal="Develop comprehensive content strategies that engage audiences",
    backstory="""You are a seasoned content strategist who:
    - Creates data-driven content plans

```

- Understands audience psychology
- Optimizes **for** multiple channels
- Aligns content **with** business goals

```
You craft strategies that convert and engage.""",  
tools=[seo_tool, social_tool],  
llm=llm,  
verbose=True,  
allow_delegation=True
```

```
)
```

```
# Agent 3: Copywriter
```

```
copywriter = Agent(  
    role="Creative Copywriter",  
    goal="Write compelling copy that converts and engages",  
    backstory="""You are an award-winning copywriter with a gift for:  
    - Persuasive writing  
    - Brand voice development  
    - Emotional storytelling  
    - A/B testing optimization
```

```
Your words drive action and build connections.""",  
tools=[seo_tool],  
llm=llm,  
verbose=True,  
allow_delegation=False
```

```
)
```

```
# Agent 4: Social Media Manager
social_media_manager = Agent(
    role="Social Media Manager",
    goal="Create engaging social media presence and grow audience",
    backstory="""You are a social media expert who:
    - Understands platform algorithms
    - Creates viral content
    - Builds communities
    - Tracks and optimizes metrics

    You turn followers into brand advocates.""",
    tools=[social_tool],
    llm=llm,
    verbose=True,
    allow_delegation=False
)
```

```
# Agent 5: SEO Specialist
seo_specialist = Agent(
    role="SEO Specialist",
    goal="Optimize all content for maximum search visibility",
    backstory="""You are an SEO expert who:
    - Stays updated on algorithm changes
    - Optimizes for user intent
    - Builds authority and rankings
    - Tracks and improves performance

    You make content discoverable and high-ranking.""",
```

```

tools=[seo_tool, research_tool],
llm=llm,
verbose=True,
allow_delegation=False
)

# Example 1: Product Launch Campaign
def product_launch_campaign():
    print("\n" + "="*60)
    print("Example 1: Product Launch Campaign")
    print("="*60 + "\n")

    # Task 1: Market Research
    research_task = Task(
        description="""Conduct comprehensive market research for our new
        AI-powered productivity tool launching next month.

        Research:
        1. Target audience demographics
        2. Market size and opportunity
        3. Competitor landscape
        4. Key differentiators
        5. Market trends

        Provide actionable insights for the campaign.""",
        agent=market_analyst,
        expected_output="Detailed market research report with recommendations"
    )

```

Task 2: Content Strategy

```
strategy_task = Task(
    description="""Based on the market research, develop a comprehensive
    content strategy for the product launch.
```

Include:

1. Key messaging pillars
2. Content calendar (4 weeks)
3. Channel strategy
4. Content types and formats
5. Success metrics

```
Align with business goals and audience needs."""
agent=content_strategist,
expected_output="Complete content strategy document"
```

```
)
```

Task 3: Copy Creation

```
copy_task = Task(
    description="""Create compelling copy assets for the launch:
```

1. Main landing page copy (hero, features, benefits, CTA)
2. Email announcement (2 versions for A/B testing)
3. Press release
4. Product tagline (5 options)

```
Follow the content strategy and emphasize key differentiators."""
```

```
    agent=copywriter,  
    expected_output="All copy assets ready for production"  
)
```

Task 4: Social Media Campaign

```
social_task = Task(  
    description="""Create a 4-week social media campaign for the launch.
```

For each week, create:

1. 5 posts for LinkedIn
2. 7 posts for Twitter/X
3. 3 posts for Instagram
4. Relevant hashtags
5. Engagement strategy

Build excitement and drive conversions."""

```
    agent=social_media_manager,  
    expected_output="Complete social media campaign calendar"  
)
```

Task 5: SEO Optimization

```
seo_task = Task(  
    description="""Optimize all content assets for SEO.
```

Optimize:

1. Landing page (meta tags, headers, content)
2. Blog posts (keywords, structure)
3. Social posts (hashtags, keywords)


```

        Provide:
        - Target keywords list
        - Optimization recommendations
        - Technical SEO checklist"",
        agent=seo_specialist,
        expected_output="SEO optimization guide and optimized content"
    )

# Create crew with sequential process
crew = Crew(
    agents=[market_analyst, content_strategist, copywriter,
            social_media_manager, seo_specialist],
    tasks=[research_task, strategy_task, copy_task, social_task, seo_task],
    process=Process.sequential,
    verbose=True
)

result = crew.kickoff()

print("\n" + "="*60)
print("CAMPAIGN COMPLETE:")
print("="*60)
print(result)

return result

```

Example 2: Hierarchical Process (Manager delegates tasks)

```

def hierarchical_campaign():
    print("\n" + "="*60)
    print("Example 2: Hierarchical Marketing Campaign")
    print("="*60 + "\n")

    # Create a manager agent
    campaign_manager = Agent(
        role="Campaign Manager",
        goal="Coordinate the team to deliver a successful marketing campaign",
        backstory="""You are an experienced campaign manager who:
        - Orchestrates complex campaigns
        - Delegates tasks effectively
        - Ensures quality and deadlines
        - Optimizes team performance

        You bring all pieces together for maximum impact.""",
        tools=[],
        llm=llm,
        verbose=True,
        allow_delegation=True
    )

    # Main task for the manager
    campaign_task = Task(
        description="""Launch a comprehensive marketing campaign for our
        Q4 product release: An AI agent platform for enterprise teams.

        Campaign objectives:

```

1. Generate 10,000 leads
2. Achieve 5% conversion rate
3. Build brand awareness
4. Establish thought leadership

Coordinate the team to deliver:

- Market analysis
- Content strategy
- All creative assets
- Social media presence
- SEO optimization

```
Ensure all elements work together cohesively.""",
agent=campaign_manager,
expected_output="Complete integrated marketing campaign"
)

# Create hierarchical crew (manager can delegate to others)
crew = Crew(
    agents=[campaign_manager, market_analyst, content_strategist,
            copywriter, social_media_manager, seo_specialist],
    tasks=[campaign_task],
    process=Process.hierarchical,
    manager_llm=llm,
    verbose=True
)

result = crew.kickoff()
```

```
print("\n" + "="*60)
print("HIERARCHICAL CAMPAIGN COMPLETE:")
print("="*60)
print(result)
```

```
return result
```

```
# Example 3: Collaborative brainstorming
```

```
def collaborative_brainstorm():
```

```
    print("\n" + "="*60)
```

```
    print("Example 3: Collaborative Brainstorming Session")
```

```
    print("="*60 + "\n")
```

```
# Task where multiple agents can contribute
```

```
brainstorm_task = Task(
```

```
    description="""Brainstorm innovative marketing ideas for reaching
    Gen Z developers with our AI agent platform.
```

```
    Consider:
```

1. Where they spend time online
2. What content resonates
3. Authentic ways to engage
4. Community-building strategies
5. Influencer partnerships

```
    Generate 10 creative, actionable campaign ideas.""",
    agent=content_strategist, # Lead agent
```

```
        expected_output="10 innovative marketing campaign concepts"
    )

# Everyone contributes perspectives
analyst_task = Task(
    description="""Provide data-driven insights about Gen Z developers:
    - Demographics and psychographics
    - Online behavior patterns
    - Platform preferences
    - Purchase decision factors""",
    agent=market_analyst,
    expected_output="Gen Z developer insights report"
)

social_task = Task(
    description="""Identify the best social platforms and tactics for
    reaching Gen Z developers. Include emerging platforms and trends.""",
    agent=social_media_manager,
    expected_output="Social platform strategy for Gen Z"
)

copy_task = Task(
    description="""Suggest messaging angles and voice that would resonate
    with Gen Z developers. What language, tone, and topics work best?""",
    agent=copywriter,
    expected_output="Messaging guidelines for Gen Z audience"
)
```

```
    crew = Crew(  
        agents=[market_analyst, content_strategist, copywriter,  
social_media_manager],  
        tasks=[analyst_task, social_task, copy_task, brainstorm_task],  
        process=Process.sequential,  
        verbose=True  
    )
```

```
    result = crew.kickoff()
```

```
    print("\n" + "="*60)  
    print("BRAINSTORM RESULTS:")  
    print("="*60)  
    print(result)
```

```
    return result
```

```
# Example 4: Crisis management
```

```
def crisis_response_team():  
    print("\n" + "="*60)  
    print("Example 4: Crisis Response Team")  
    print("="*60 + "\n")
```

```
# Simulate a PR crisis scenario
```

```
crisis_assessment = Task(  
    description="""A major tech blog published a critical article about  
our AI agent platform's data privacy practices.
```

Assess:

1. Severity of the situation
 2. Key concerns raised
 3. Potential impact on brand
 4. Stakeholder reactions
 5. Immediate actions needed""",
- agent=market_analyst,
expected_output="Crisis assessment and recommendations"

)

response_strategy = Task(
description=""Develop a comprehensive crisis response strategy:

1. Official statement positioning
 2. Communication channels to use
 3. Stakeholder outreach plan
 4. Timeline for responses
 5. Long-term reputation recovery""",
- agent=content_strategist,
expected_output="Crisis response strategy"

)

statement_draft = Task(
description=""Write the official company statement addressing the concerns.

The statement must:

- Acknowledge concerns seriously
- Provide clear facts

- Show commitment to privacy
- Rebuild trust
- Call **for** dialogue

```
Tone: Transparent, professional, empathetic""",  
agent=copywriter,  
expected_output="Official crisis response statement"
```

```
)
```

```
social_response = Task(  
    description="""Create social media response plan:
```

1. Initial acknowledgment posts
2. Ongoing communication strategy
3. Community management guidelines
4. FAQ responses
5. Monitoring plan

```
Focus on transparency and engagement.""",  
agent=social_media_manager,  
expected_output="Social media crisis response plan"
```

```
)
```

```
crew = Crew(  
    agents=[market_analyst, content_strategist, copywriter,  
social_media_manager],  
    tasks=[crisis_assessment, response_strategy, statement_draft,  
social_response],
```



```
        process=Process.sequential,  
        verbose=True  
    )
```

```
result = crew.kickoff()
```

```
print("\n" + "="*60)  
print("CRISIS RESPONSE PLAN:")  
print("="*60)  
print(result)
```

```
return result
```

```
# Run examples
```

```
if __name__ == "__main__":  
    print("CrewAI Multi-Agent Examples - Marketing Team")  
    print("=" * 60)
```

```
# Uncomment the example you want to run:
```

```
product_launch_campaign()  
# hierarchical_campaign()  
# collaborative_brainstorm()  
# crisis_response_team()
```

```
print("\n" + "="*60)  
print("Marketing campaign completed!")  
print("="*60)
```

